

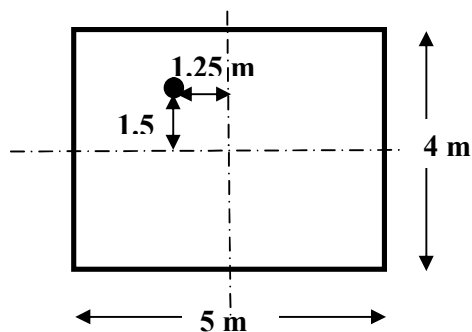
CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fourth Semester of B.Tech.Theory Examination (CIVIL)****APRIL – MAY 2018****CL 246 STRUCTURAL ANALYSIS- I****Date: 04.05.2018 (Friday)****Time: 10:00 a.m. to 01:00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat sketch wherever required.
4. Use of scientific calculator is allowed.

SECTION - I**Q - 1 Do as directed.**

- i) Distinguish between direct and bending stress by means of a diagram. [01]
- ii) Find the greatest eccentricity which a load can have without producing tension [02]
in a short hollow circular column having outside diameter 'D' and inside diameter 'd'.
- iii) If the crushing stress in the material of a mild steel column is 200 MPa and [01]
modulus of elasticity is 200 GPa then Euler's formula for critical load is
applicable for a slenderness ratio equal to or greater than
(a) 100 (b) 50 (c) 60 (d) 80
- iv) Classify the columns based on slenderness ratio. [01]
- v) The effective length of a long column of length 'L', fixed at the bottom and free [01]
at the top, is _____.
(a) 2L (b) L (c) L/2 (d) 3L

- Q - 2 (a)** i) What is the limitation of Euler's formula? Explain with example. [02]
- ii) A column 5 m x 4 m is subjected to an eccentric load of 50 kN as shown in [05]
figure 1. Calculate maximum and minimum intensity of stresses in the column.

**Figure 1 [Q-2 (a)(ii)]****OR**

Q - 2 (a) A masonry wall, 6 m high, is of solid rectangular section, 4 m wide and 1.5 m thick. A horizontal wind pressure of 1 kN/m^2 acts on the width of the wall side. Find the maximum and minimum stresses induced on the base, if the unit weight of masonry is 22.4 kN/m^3 . **[07]**

Q - 2 (b) i) Justify the statement “With the help of Rankine’s formula we see that safe axial load is reduced by the factor $\frac{1}{1+a\lambda^2}$ “. **[02]**

ii) Compare the strength of solid circular column section of diameter 200 mm and hollow circular column section of same cross sectional area and thickness 30 mm. The other parameter are same for both the sections. **[05]**

OR

Q - 2 (b) i) What are the assumptions made in the Euler’s theory? **[03]**

ii) Derive the Euler’s crippling formula for a column of length ‘L’ with one end fixed and other end free. **[04]**

Q - 3 (a) Answer any two questions. **[10]**

i) Draw qualitative influence line diagram for reaction at A, reaction at B, shear at section 4 m from left support and bending at section 4 m from left support for the over hanging beam of span 10 m. Consider the span between support A and support B as 7 m and over hang portion span between support B and free end as 3 m. Also show the ordinates at important location.

ii) Two wheel loads of 12 kN and 20 kN have fixed distance of 2 m between them. 12 kN load is the leading load. The wheels cross a beam of span 8m from left to right. Draw the influence line for shear force and bending moment for a point 3 m from left support. Also find maximum values of shear force and bending moment at section.

iii) A udl of intensity 80 kN/m and 30 m length crosses a girder of 25 m span. Calculate maximum values of shear force and bending moment in the girder.

Q - 3 (b) A shaft has to transmit 10 KW power at a speed of 15 Hz. If maximum shear stress should not exceed 60 MPa, find the suitable diameter for solid circular section. **[05]**

SECTION – II

Q - 4 Do as directed.

i) State the equation of Torsion. **[01]**

ii) What is angle of twist? **[01]**

iii) In a plane truss member, there are 2 degrees of freedom per joint. True or False? **[01]**

- iv) Calculate Static Indeterminacy and Kinematic Indeterminacy for the following [02]
beam. **Refer figure 2.**

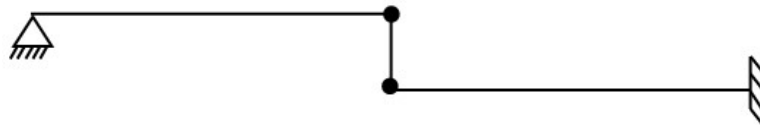


Figure 2 [Q-4 (iv)]

- v) Calculate Static Indeterminacy and Kinematic Indeterminacy for the following [02]
plane frame. **Refer figure 3.**

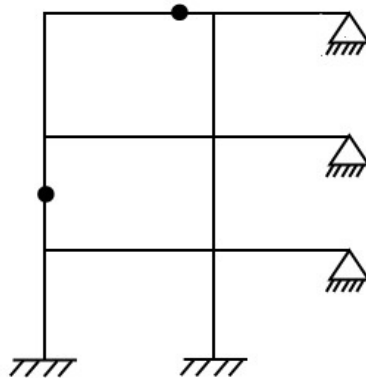


Figure 3 [Q-4 (v)]

Q - 5 (a) Explain longitudinal stress and circumferential stress. [02]

Q - 5 (b) Calculate slope at 'C' i.e. θ_C and deflection at 'C' i.e. δ_C for the cantilever beam shown in **figure 4** using Moment area method **OR** Conjugate beam method. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 3 \times 10^8 \text{ mm}^4$. [06]

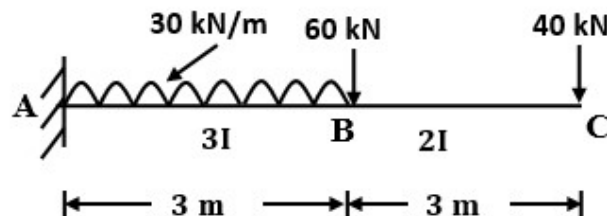


Figure 4 [Q-5 (b)]

Q - 5 (c) Calculate deflection at 'C' i.e. δ_C and deflection at 'D' i.e. δ_D for the beam shown in **figure 5** using Double integration method **OR** Macaulay's method. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 3 \times 10^8 \text{ mm}^4$. [06]

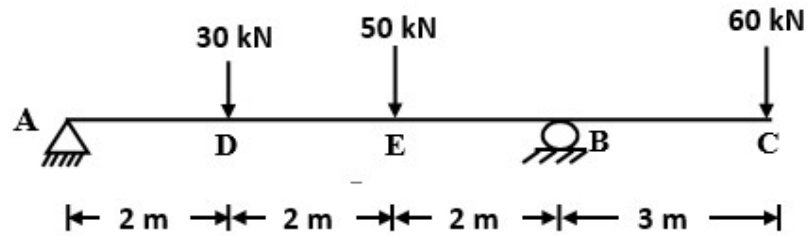


Figure 5 [Q-5 (c)]

- Q - 6 (a)** A closed cylindrical vessel made of steel plates 6 mm thick with plane ends carries fluid under a pressure of 5 N/mm^2 . The diameter of the cylinder is 300 mm and the length is 1000 mm. Calculate the longitudinal and circumferential stresses in the cylinder wall. Also determine the change in diameter, length and volume of the cylinder. Consider $E = 2.1 \times 10^5 \text{ N/mm}^2$ and $\mu = 0.3$. [05]
- Q - 6 (b)** A 3 hinged circular arch of 20 m span and 4 m of central rise is loaded with a concentrated force of 12 kN at 6 m from the left hinge. Find (i) Support reactions (ii) Maximum positive bending moment in left span and maximum negative bending moment in right span. [09]

OR

- Q - 6 (b)** A three hinged parabolic arch has span 12 m and central rise of 2.5 m. It carries uniformly distributed load of 30 kN/m over left half of the span. Calculate the reactions at both ends. Also calculate the bending moment, normal thrust and radial shear at 3 m from left hinge. [09]
